

EDUCATION

Indian Institute of Technology, Kanpur

Bachelor of Technology in Materials Science and Engineering

2020–2024

WORK EXPERIENCE

ixigo

Software Engineer 2, AI Products

Aug'24 – Present

Lead end-to-end TARA engineering

across agent harness, provider-neutral orchestration, LiveKit/WebRTC, client actions, evals, and observability

- Architected **TARA** as ixigo's **embedded multimodal conversational AI platform and event-driven SDK**, decomposing real-time media, orchestration, provider adapters, native-client actions, and observability into independently evolvable layers
- Engineered the **LiveKit/WebRTC transport** for bidirectional audio, participant lifecycle, and ordered data channels, integrating a composable, provider-neutral Pipecat pipeline spanning VAD/STT, semantic turn-taking, LLM/TTS, and policy-driven barge-in
- Architected a **contextual agent harness** that compiles journey state, session memory, and live context into stage-scoped prompts and toolsets; built ixigo's **internal MCP server** for flight and hotel discovery-to-booking workflows as reusable agent capabilities
- Scaled TARA to **60,000+ in-app voice sessions/day**, sustaining **1.5 s p95** from detected end-of-turn to first assistant audio while cutting **infrastructure cost per voice minute by approximately 50%** through backpressure and overload protection
- Architected **Agent Studio** for agent observability and evaluation, combining trace reconstruction, LLM-as-judge workflows, regression detection, failure clustering, and human-gated, backtested evaluator refinement against labeled and golden-set data for release gates
- Designed its **revision-aware ClickHouse observability data model and Kafka/Avro telemetry contract** for high-volume agent traces, defining event-time and revision semantics, deduplication, governed retention, and cost/latency rollups for operational analytics
- Architected and operated a distinct multi-agent conversational telephony platform handling **35,000+ calls/day**, owning orchestration, concurrency controls, capacity planning, failure recovery, and fleet-wide reliability under sustained production traffic and peak demand
- Built **ixi-skill auto-refiner**, an autoresearch-inspired evaluation loop adopted organization-wide to improve internal agent skills

Overlay AI

Founding Engineer

Apr'24 – Aug'24

- Joined as first engineer and built the core **LLM/RAG product** end to end, helping onboard the company's first paying customers
- Owned AWS infrastructure, CI/CD automation, and production operations across the startup's microservice platform
- Built authenticated WebSocket messaging and ClickHouse-backed analytics, extending Elixir/Phoenix services for product telemetry

OPEN SOURCE & SELECTED PROJECTS

Open-Source Contributions

Jul'26 – Present

- Contributed multiple fixes across **LiveKit**, **Pipecat**, and **PR-Agent** spanning real-time voice reliability, SDK correctness, provider integrations, and regression coverage, with several changes merged upstream

AgentRelay

Apr'26 – Present

- Created an **open-source coordination layer** enabling independently owned Codex and Claude agents to exchange authenticated, typed handoffs across repositories and machines while preserving separate credentials, source boundaries, and execution authority
- Designed a durable **Mission control plane** with ordered delivery, leases, monotonic fencing, exact replay, bounded retries, crash recovery, and revocation to preserve task authority, delivery safety, and recoverability across partial failures and process restarts
- Published an **MCP mailbox** with identities, provenance, trust controls, blocking, and audit trails, defining scoped authority across relay and host layers while preserving local credentials, execution authority, and durable handoff state across asynchronous sessions

Murmur – Multimodal Voice Tutor

Dec'25 – Present

- Built a voice-first tutor combining interruptible WebRTC dialogue with a canvas for diagrams, equations, and explanations in real time
- Built **four-layer memory** spanning short-term context, episodic summaries, semantic memory, and persistent, evolving learner profiles
- Unified WebRTC voice and SSE chat through shared agent, memory, tool, and canvas layers with pluggable providers and shared state

cmux Agent Orchestrator

Aug'26 – Present

- Released an Obsidian cockpit linking cmux-owned Claude Code/Codex sessions to durable tasks and one-click terminal focus
- Built a typed cmux boundary with canonical UUIDs, bounded in-memory evidence, conservative agent detection, and fail-closed actions

ACHIEVEMENTS

- Inter IIT Tech Meet 10.0, 2022:** Silver Medal on ISRO's X-ray burst automation among 23 IITs; IIT Kanpur placed 2nd overall
- Google Hash Code, 2021:** Secured **Global Rank 5,401** among 100K+ participants in Google's team programming competition
- ixigo, 2025:** Hackweek runner-up among 200+ developers for Reel2Trip; Premio Award recipient for engineering contribution

LEADERSHIP

Head Web & App, Techkriti'23 • Web Secretary, Hall Executive Committee (Hall 3)

Led web/app delivery for Techkriti'23's 400-member team; shipped 10+ portals for **14.5K+ users**, **350K monthly** and **29.5K peak-day views**

Jul'22 – May'23

EARLIER EXPERIENCE

Software Engineering Internships

FischerJordan (PPO offered), ARTPARK @ IISc, Llama Mindfulness, Vaticanari Solutions, and C3iHub, IIT Kanpur

2021–2024

Languages & Backend: Python, TypeScript, JavaScript, Elixir, C++; FastAPI, Node.js, WebSockets, SSE, RPC

Agent & Realtime AI: Pipecat, LiveKit, WebRTC, MCP, LLM/RAG & Multi-Agent Orchestration, Evaluation, Observability

Data & Infrastructure: ClickHouse, Kafka/Avro, Grafana/Loki, Valkey/Redis, PostgreSQL, Docker, Kubernetes, AWS, CI/CD